

#2 Quiz Chapter 03, 04&05

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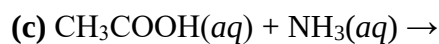
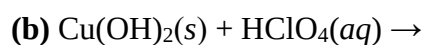
1-5 ADBAE

6-10 CCAAA

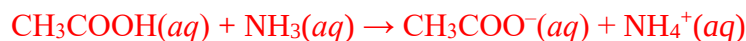
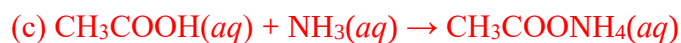
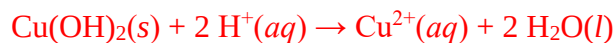
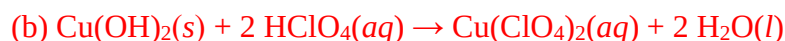
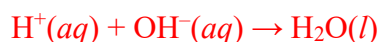
11-15 EEAAB

16-17 EC

18. (6 points) Complete and balance the following molecular equations, and then write the net ionic equation for each:



Answer:



19. (3 points) Which of the following solutions is the most basic: (a) 0.100 M NH_3 ; (b) 0.150 M KOH; (c) 0.100 M $\text{Ba}(\text{OH})_2$? Explain why.

Answer:

$\text{NH}_3(aq)$ is a weak base, while KOH and $\text{Ba}(\text{OH})_2$ are strong bases.

$$[\text{OH}^-]_{\text{Ba}(\text{OH})_2} = 0.200\text{ M}$$

$$[\text{OH}^-]_{\text{KOH}} = 0.150\text{ M}$$

$$[\text{OH}^-]_{\text{NH}_3} < 0.100\text{ M}$$

Thus, 0.100 M $\text{Ba}(\text{OH})_2$ is the most basic.

20. (3 points) You want to analyze a cadmium nitrate solution. What mass of NaOH

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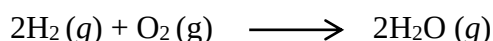
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is needed to precipitate the Cd^{2+} ions from 35.0 mL of 0.500 M $\text{Cd}(\text{NO}_3)_2$ solution.

Answer:

Number of moles of Cd^{2+} ions in 35.0 mL of 0.500 M cadmium nitrate = $(0.500 \text{ mol}/1000 \text{ mL}) \times 35.0 \text{ mL} = 0.0175 \text{ moles}$, one mole of Cd^{2+} require 2 moles of NaOH, therefore 0.0175 moles will require **0.0350 moles of NaOH**. The molar mass of NaOH = 40.01 g/mol, 0.0350 moles of NaOH = $0.0350 \text{ mol} \times 40.01 \text{ g/mol} = \mathbf{1.40 \text{ g}}$

21. (6 points) The reaction



is used to produce electricity in a hydrogen fuel cell. Suppose a fuel cell contains 150 g of $\text{H}_2(\text{g})$ and 1500 g of $\text{O}_2(\text{g})$ (each measured to two significant figures). How many grams of water can form?

From the balanced equation, we have the stoichiometric relations:



Using the molar mass of each substance, we calculate the number of moles of each reactant:

$$\text{Moles H}_2 = (150 \text{ g H}_2) \left(\frac{1 \text{ mol H}_2}{2.02 \text{ g H}_2} \right) = 74 \text{ mol H}_2$$

$$\text{Moles O}_2 = (1500 \text{ g O}_2) \left(\frac{1 \text{ mol O}_2}{32.0 \text{ g O}_2} \right) = 47 \text{ mol O}_2$$

The coefficients in the balanced equation indicate that the reaction requires 2 mol of H_2 for every 1 mol of O_2 . Therefore, for all the O_2 to completely react, we would need $2 \times 47 = 94 \text{ mol}$ of H_2 . Since there are only 74 mol of H_2 , all of the O_2 cannot react, so it is the excess reactant, and H_2 must be the limiting reactant. (Notice that the limiting reactant is not necessarily the one present in the lowest amount.)

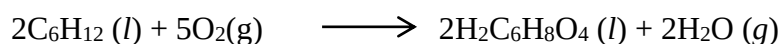
We use the given quantity of H_2 (the limiting reactant) to calculate the quantity of water formed. We could begin this calculation with the given H_2 mass, 150 g, but we can save a step by starting with the moles of H_2 , 74 mol, we just calculated:

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$$\begin{aligned}\text{Grams H}_2\text{O} &= (74 \cancel{\text{ mol H}_2}) \left(\frac{2 \cancel{\text{ mol H}_2\text{O}}}{2 \cancel{\text{ mol H}_2}} \right) \left(\frac{18.0 \text{ g H}_2\text{O}}{1 \cancel{\text{ mol H}_2\text{O}}} \right) \\ &= 1.3 \times 10^2 \text{ g H}_2\text{O}\end{aligned}$$

22. (6 points) Adipic acid, $\text{H}_2\text{C}_6\text{H}_8\text{O}_4$, used to produce nylon, is made commercially by a reaction between cyclohexane C_6H_{12} and O_2 :



(a) Assume that you carry out this reaction with 25.0 g of cyclohexane and that cyclohexane is the limiting reactant. What is the theoretical yield of adipic acid?

(b) If you obtain 33.5 g of adipic acid, what is the percent yield for the reaction?

(a) The theoretical yield is

$$\begin{aligned}\text{Grams H}_2\text{C}_6\text{H}_8\text{O}_4 &= (25.0 \cancel{\text{ g C}_6\text{H}_{12}}) \left(\frac{1 \cancel{\text{ mol C}_6\text{H}_{12}}}{84.0 \cancel{\text{ g C}_6\text{H}_{12}}} \right) \left(\frac{2 \cancel{\text{ mol H}_2\text{C}_6\text{H}_8\text{O}_4}}{2 \cancel{\text{ mol C}_6\text{H}_{12}}} \right) \left(\frac{146.0 \text{ g H}_2\text{C}_6\text{H}_8\text{O}_4}{1 \cancel{\text{ mol H}_2\text{C}_6\text{H}_8\text{O}_4}} \right) \\ &= 43.5 \text{ g H}_2\text{C}_6\text{H}_8\text{O}_4\end{aligned}$$

$$\text{(b) Percent yield} = \frac{\text{actual yield}}{\text{theoretical yield}} \times 100\% = \frac{33.5 \text{ g}}{43.5 \text{ g}} \times 100\% = 77.0\%$$

23. (5 points) Classify these dissolved substances as a strong electrolyte, weak electrolyte, or nonelectrolyte: CaCl_2 , HNO_3 , $\text{C}_2\text{H}_5\text{OH}$ (ethanol), HCOOH (formic acid), KOH .

Answer:

Strong electrolyte: CaCl_2 , HNO_3 , KOH

Weak electrolyte: HCOOH

Nonelectrolyte: $\text{C}_2\text{H}_5\text{OH}$

24. (5 points) Determine the oxidation number of sulfur in (a) H_2S , (b) S_8 , (c) SCl_2 , (d) Na_2SO_3 , (e) SO_4^{2-}

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Answer:

(a) H_2S , -2

(b) S_8 , 0

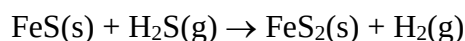
(c) SCl_2 , +2

(d) Na_2SO_3 , +4

(e) SO_4^{2-} , +6

25. (4 points) Write a redox chemical equation that could happen in the deep-sea hydrothermal vent. And label the oxidation state of each element.

Answer:



FeS: +2

FeS: -2

H_2S : +1

H_2S : -2

FeS₂: +4

FeS₂: -2

H_2 : 0

26. (6 points) When a student mixes 50 mL of 1.0 M HCl and 50 mL of 1.0 M NaOH in a coffee-cup calorimeter, the temperature of the resultant solution increases from 21.0 °C to 27.5 °C. Calculate the enthalpy change for the reaction in kJ/mol HCl, assuming that the calorimeter loses only a negligible quantity of heat, that the total volume of the solution is 100 mL, that its density is 1.0 g/mL, and that its specific heat is 4.18 J/g·K.

Solve

Because the total volume of the solution is 100 mL, its mass is

$$(100 \text{ mL})(1.0 \text{ g/mL}) = 100 \text{ g}$$

The temperature change is

$$\Delta T = 27.5^\circ\text{C} - 21.0^\circ\text{C} = 6.5^\circ\text{C} = 6.5 \text{ K}$$

Using Equation 5.23, we have

$$\begin{aligned} q_{\text{rxn}} &= -C_s \times m \times \Delta T \\ &= -(4.18 \text{ J/g} \cdot \text{K})(100 \text{ g})(6.5 \text{ K}) = -2.7 \times 10^3 \text{ J} = -2.7 \text{ kJ} \end{aligned}$$

Because the process occurs at constant pressure,

$$\Delta H = q_p = -2.7 \text{ kJ}$$

To express the enthalpy change on a molar basis, we use the fact that the number of moles of HCl is given by the product of the volume (50 mL = 0.050 L) and concentration (1.0 M = 1.0 mol/L) of the HCl solution:

$$(0.050 \text{ L})(1.0 \text{ mol/L}) = 0.050 \text{ mol}$$

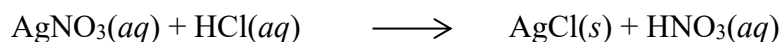
Thus, the enthalpy change per mole of HCl is

$$\Delta H = -2.7 \text{ kJ}/0.050 \text{ mol} = -54 \text{ kJ/mol}$$

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27. **(5 points)** When 50.0 mL of 0.100 M AgNO₃ and 50.0 mL of 0.100 M HCl are mixed in a constant-pressure calorimeter, the temperature of the mixture increases from 22.30 °C to 23.11 °C. The temperature increase is caused by the following reaction:



Calculate ΔH for this reaction in AgNO₃, assuming that the combined solution has a mass of 100.0 g and a specific heat of 4.18 J/g °C.

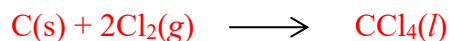
$$\Delta T = 23.11^\circ\text{C} - 22.30^\circ\text{C} = 0.81^\circ\text{C}$$

$$Q = -C_s \times m \times \Delta T = -338.58 \text{ J}$$

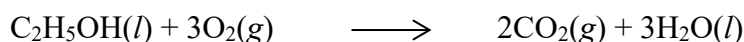
$$n = 50 \text{ mL} \times 0.100 \text{ M} = 0.005 \text{ mol}$$

$$\Delta H = \frac{Q}{n} = -67716 \text{ J} \cdot \text{mol}^{-1} = -67.72 \text{ kJ} \cdot \text{mol}^{-1}$$

28. **(2 points)** Write the equation corresponding to the standard enthalpy of formation of liquid carbon tetrachloride (CCl₄).



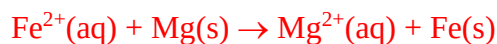
29. **(4 points)** Use the following Table (on the Page 11) to calculate the enthalpy change for the combustion of 1 mol of ethanol:



$$\Delta H = 2\Delta H_f^\circ(\text{CO}_2) + 3\Delta H_f^\circ(\text{H}_2\text{O}) - \Delta H_f^\circ(\text{C}_2\text{H}_5\text{OH}) = -1366.7 \text{ kJ} \cdot \text{mol}^{-1}$$

30. **(2 points)** Will an aqueous solution of iron(II) chloride oxidize magnesium metal? If so, write the balanced molecular and net ionic equations for the reaction.

Answer:



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31. (3 points) Calculate the molarity of a solution made by dissolving 23.4 g of sodium sulfate (Na_2SO_4) in enough water to form 125 mL of solution.

Solve The number of moles of Na_2SO_4 is obtained by using its molar mass:

$$\text{Moles Na}_2\text{SO}_4 = (23.4 \text{ g Na}_2\text{SO}_4) \left(\frac{1 \text{ mol Na}_2\text{SO}_4}{142.1 \text{ g Na}_2\text{SO}_4} \right) = 0.165 \text{ mol Na}_2\text{SO}_4$$

Converting the volume of the solution to liters: $\text{Liters soln} = (125 \text{ mL}) \left(\frac{1 \text{ L}}{1000 \text{ mL}} \right) = 0.125 \text{ L}$

Thus, the molarity is $\text{Molarity} = \frac{0.165 \text{ mol Na}_2\text{SO}_4}{0.125 \text{ L soln}} = 1.32 \frac{\text{mol Na}_2\text{SO}_4}{\text{L soln}} = 1.32 \text{ M}$

32. (4 points) How many milliliters of 3.0 M H_2SO_4 are needed to make 450 mL of 0.10 M H_2SO_4 ?

Solve Calculating the moles of H_2SO_4 in the dilute solution:

$$\begin{aligned} \text{Moles H}_2\text{SO}_4 \text{ in dilute solution} &= (0.450 \text{ L soln}) \left(\frac{0.10 \text{ mol H}_2\text{SO}_4}{1 \text{ L soln}} \right) \\ &= 0.045 \text{ mol H}_2\text{SO}_4 \end{aligned}$$

Calculating the volume of the concentrated solution that contains 0.045 mol H_2SO_4 :

$$\text{L conc soln} = (0.045 \text{ mol H}_2\text{SO}_4) \left(\frac{1 \text{ L soln}}{3.0 \text{ mol H}_2\text{SO}_4} \right) = 0.015 \text{ L soln}$$

Converting liters to milliliters gives 15 mL.

If we apply Equation 4.34, we get the same result:

$$\begin{aligned} (3.0 \text{ M})(V_{\text{conc}}) &= (0.10 \text{ M})(450 \text{ mL}) \\ (V_{\text{conc}}) &= \frac{(0.10 \text{ M})(450 \text{ mL})}{3.0 \text{ M}} = 15 \text{ mL} \end{aligned}$$

33. (5 points) One commercial method used to peel potatoes is to soak them in a NaOH solution for a short time and then remove the potatoes and spray off the peel. The NaOH concentration is normally 3 to 6 M, and the solution must be analyzed

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periodically. In one such analysis, 45.7 mL of 0.500 M H_2SO_4 is required to neutralize 20.0 mL of NaOH solution. What is the concentration of the NaOH solution?

$$\begin{aligned}\text{Moles } \text{H}_2\text{SO}_4 &= (45.7 \text{ mL soln}) \left(\frac{1 \text{ L soln}}{1000 \text{ mL soln}} \right) \left(\frac{0.500 \text{ mol } \text{H}_2\text{SO}_4}{1 \text{ L soln}} \right) \\ &= 2.28 \times 10^{-2} \text{ mol } \text{H}_2\text{SO}_4\end{aligned}$$

Acids react with metal hydroxides to form water and a salt. Thus, the balanced equation for the neutralization reaction is



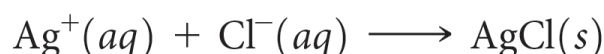
According to the balanced equation, $1 \text{ mol } \text{H}_2\text{SO}_4 \approx 2 \text{ mol NaOH}$. Therefore,

$$\begin{aligned}\text{Moles NaOH} &= (2.28 \times 10^{-2} \text{ mol } \text{H}_2\text{SO}_4) \left(\frac{2 \text{ mol NaOH}}{1 \text{ mol } \text{H}_2\text{SO}_4} \right) \\ &= 4.56 \times 10^{-2} \text{ mol NaOH}\end{aligned}$$

Knowing the number of moles of NaOH in 20.0 mL of solution allows us to calculate the molarity of this solution:

$$\begin{aligned}\text{Molarity NaOH} &= \frac{\text{mol NaOH}}{\text{L soln}} \\ &= \left(\frac{4.56 \times 10^{-2} \text{ mol NaOH}}{20.0 \text{ mL soln}} \right) \left(\frac{1000 \text{ mL soln}}{1 \text{ L soln}} \right) \\ &= 2.28 \frac{\text{mol NaOH}}{\text{L soln}} = 2.28 \text{ M}\end{aligned}$$

34. (5 points) The quantity of Cl^- in a municipal water supply is determined by titrating the sample with Ag^+ . The precipitation reaction taking place during the titration is



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The end point in this type of titration is marked by a change in color of a special type of indicator.

(a) How many grams of chloride ion are in a sample of the water if 20.2 mL of 0.100 M Ag^+ is needed to react with all the chloride in the sample? (b) If the sample has a mass of 10.0 g, what percentage of Cl^- does it contain?

(a)

Solve

$$\begin{aligned}\text{Moles } \text{Ag}^+ &= (20.2 \text{ mL soln}) \left(\frac{1 \text{ L soln}}{1000 \text{ mL soln}} \right) \left(\frac{0.100 \text{ mol Ag}^+}{\text{L soln}} \right) \\ &= 2.02 \times 10^{-3} \text{ mol Ag}^+\end{aligned}$$

From the balanced equation we see that $1 \text{ mol Ag}^+ \approx 1 \text{ mol Cl}^-$.

Using this information and the molar mass of Cl, we have

$$\begin{aligned}\text{Grams } \text{Cl}^- &= (2.02 \times 10^{-3} \text{ mol Ag}^+) \left(\frac{1 \text{ mol Cl}^-}{1 \text{ mol Ag}^+} \right) \left(\frac{35.5 \text{ g Cl}^-}{\text{mol Cl}^-} \right) \\ &= 7.17 \times 10^{-2} \text{ g Cl}^-\end{aligned}$$

(b)

Solve

$$\text{Percent } \text{Cl}^- = \frac{7.17 \times 10^{-2} \text{ g}}{10.0 \text{ g}} \times 100\% = 0.717\% \text{ Cl}^-$$

35. (6 points) A sample of 70.5 mg of potassium phosphate is added to 15.0 mL of 0.050 M silver nitrate, resulting in the formation of a precipitate. (a) Write the molecular equation for the reaction. (b) What is the limiting reactant in the reaction? (c) Calculate the theoretical yield, in grams, of the precipitate that forms.

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SOLUTION

- (a) Potassium phosphate and silver nitrate are both ionic compounds. Potassium phosphate contains K^+ and PO_4^{3-} ions, so its chemical formula is K_3PO_4 . Silver nitrate contains Ag^+ and NO_3^- ions, so its chemical formula is $AgNO_3$. Because both reactants are strong electrolytes, the solution contains, K^+ , PO_4^{3-} , Ag^+ , and NO_3^- ions before the reaction occurs. According to the solubility guidelines in Table 4.1, Ag^+ and PO_4^{3-} form an insoluble compound, so Ag_3PO_4 will precipitate from the solution. In contrast, K^+ and NO_3^- will remain in solution because KNO_3 is water soluble. Thus, the balanced molecular equation for the reaction is



- (b) To determine the limiting reactant, we must examine the number of moles of each reactant. ∞ (Section 3.7) The number of moles of K_3PO_4 is calculated from the mass of the sample using the molar mass as a conversion factor. ∞ (Section 3.4) The molar mass of K_3PO_4 is $3(39.1) + 31.0 + 4(16.0) = 212.3$ g/mol. Converting milligrams to grams and then to moles, we have

$$(70.5 \text{ mg } K_3PO_4) \left(\frac{10^{-3} \text{ g } K_3PO_4}{1 \text{ mg } K_3PO_4} \right) \left(\frac{1 \text{ mol } K_3PO_4}{212.3 \text{ g } K_3PO_4} \right) = 3.32 \times 10^{-4} \text{ mol } K_3PO_4$$

We determine the number of moles of $AgNO_3$ from the volume and molarity of the solution. (Section 4.5) Converting milliliters to liters and then to moles, we have

$$(15.0 \text{ mL}) \left(\frac{10^{-3} \text{ L}}{1 \text{ mL}} \right) \left(\frac{0.050 \text{ mol } AgNO_3}{\text{L}} \right) = 7.5 \times 10^{-4} \text{ mol } AgNO_3$$

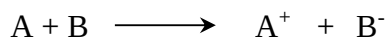
Comparing the amounts of the two reactants, we find that there are $(7.5 \times 10^{-4}) / (3.32 \times 10^{-4}) = 2.3$ times as many moles of $AgNO_3$ as there are moles of K_3PO_4 . According to the balanced equation, however, 1 mol K_3PO_4 requires 3 mol $AgNO_3$. Thus, there is insufficient $AgNO_3$ to consume the K_3PO_4 , and $AgNO_3$ is the limiting reactant.

- (c) The precipitate is Ag_3PO_4 , whose molar mass is $3(107.9) + 31.0 + 4(16.0) = 418.7$ g/mol. To calculate the number of grams of Ag_3PO_4 that could be produced in this reaction (the theoretical yield), we use the number of moles of the limiting reactant, converting $\text{mol } AgNO_3 \Rightarrow \text{mol } Ag_3PO_4 \Rightarrow \text{g } Ag_3PO_4$. We use the coefficients in the balanced equation to convert moles of $AgNO_3$ to moles Ag_3PO_4 , and we use the molar mass of Ag_3PO_4 to convert the number of moles of this substance to grams.

$$(7.5 \times 10^{-4} \text{ mol } AgNO_3) \left(\frac{1 \text{ mol } Ag_3PO_4}{3 \text{ mol } AgNO_3} \right) \left(\frac{418.7 \text{ g } Ag_3PO_4}{1 \text{ mol } Ag_3PO_4} \right) = 0.10 \text{ g } Ag_3PO_4$$

The answer has only two significant figures because the quantity of $AgNO_3$ is given to only two significant figures.

36. (3 points) In the following redox reaction, the oxidation states of reactant A and B are both zero. Please fill the blank with A or B:



- 1) The reactant _____ is an oxidant;
- 2) The reactant _____ is oxidized;
- 3) The reaction of reactant _____ is oxidation reaction.

(1) B (2) A (3) A